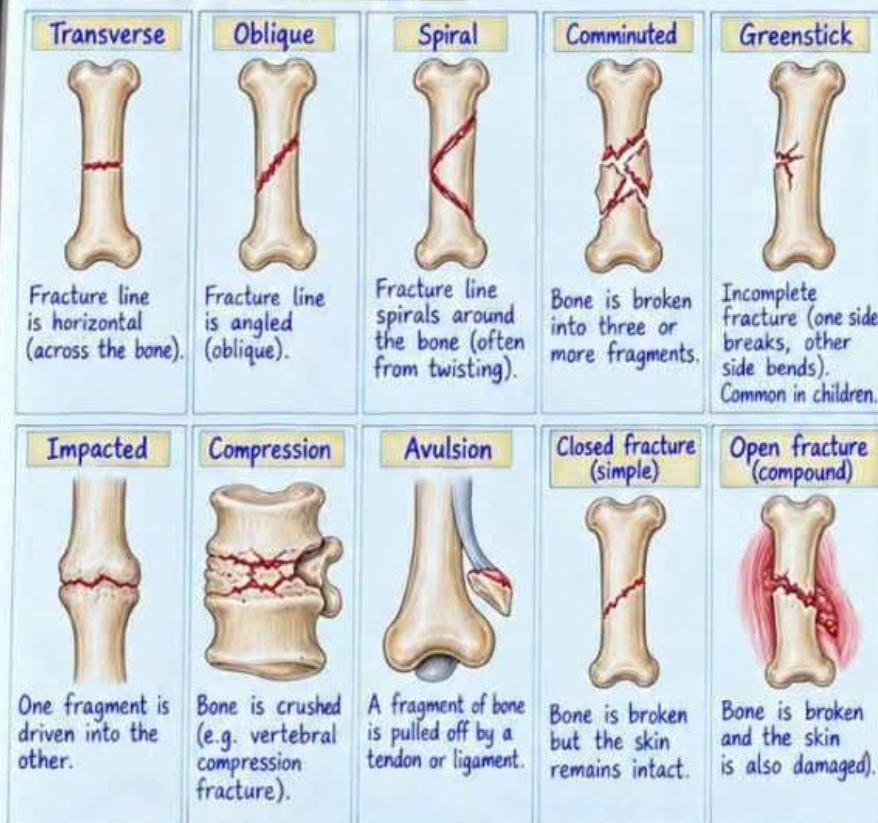
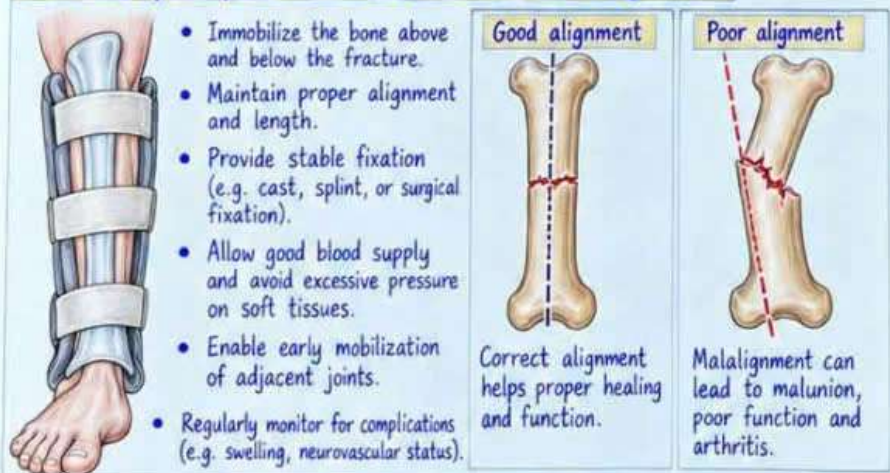


FRACTURE TYPES AND BONE HEALING

A. Common fracture types



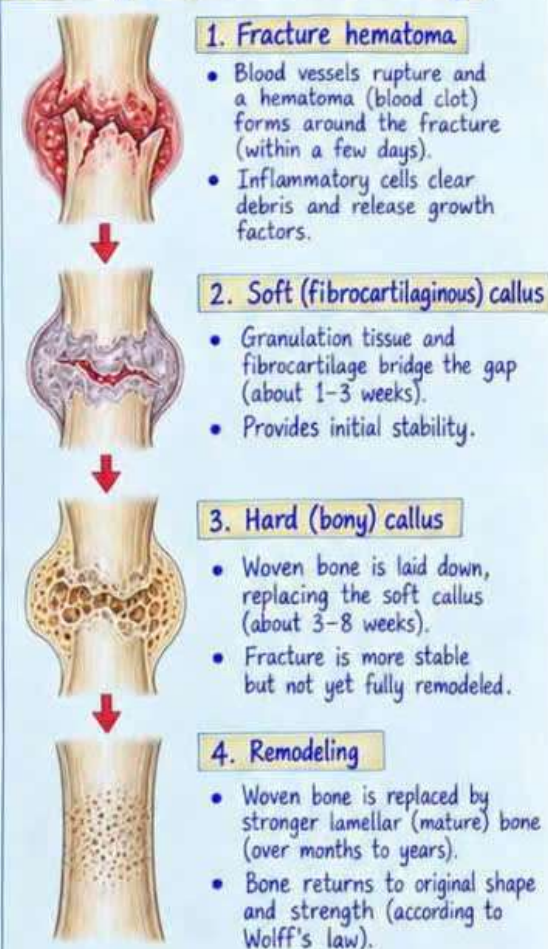
C. Key principles of immobilization and alignment



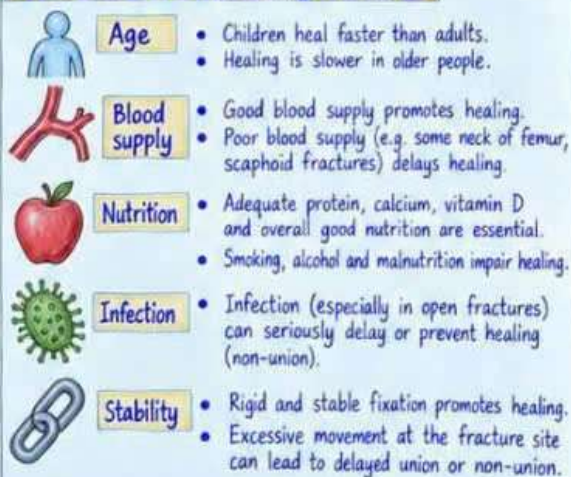
E. Key points / summary

- Fractures can be complete or incomplete, and have different patterns (transverse, oblique, spiral, comminuted, greenstick, etc.).
- Bone healing occurs in stages: hematoma → soft callus → hard callus → remodeling.
- Proper immobilization, alignment and stable fixation are essential for good healing.
- Healing is influenced by age, blood supply, nutrition, infection and stability.
- Complications include delayed union, non-union, malunion and infection.
- With correct management, most fractures heal well and bone can regain near-normal strength and function.

B. Stages of fracture healing



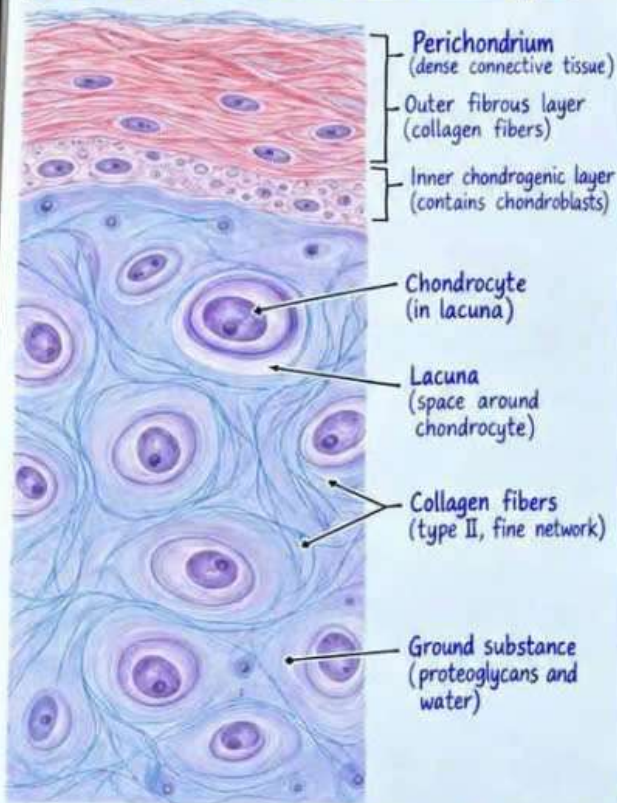
D. Factors affecting healing



Healthy bones heal - with time, care and the right conditions! ♥

CARTILAGE ANATOMY

A. General structure of cartilage



B. Hyaline cartilage



Structure:

- Glassy, smooth matrix rich in type II collagen (fine fibers).
- Chondrocytes in lacunae, often in small groups (isogenous groups).

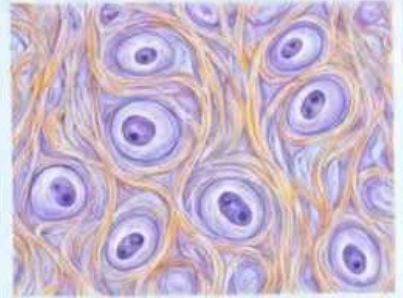
Function:

- Provides support with flexibility.
- Reduces friction at joints.
- Resists compression.

Locations:

- Articular surfaces of synovial joints.
- Costal cartilages (ribs).
- Nose, trachea, larynx (e.g. thyroid cartilage).
- Fetal skeleton (temporary).

C. Elastic cartilage



Structure:

- Similar to hyaline cartilage but with abundant elastic fibers (in addition to type II collagen).
- Chondrocytes in lacunae.

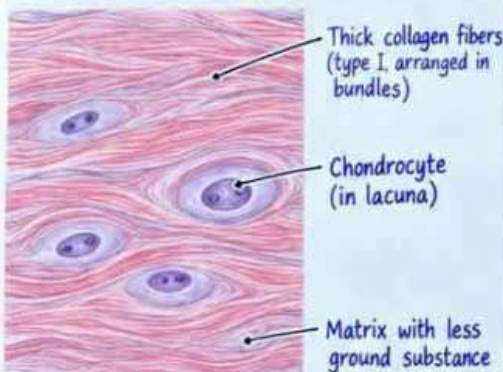
Function:

- Flexible and maintains shape after bending.
- Provides elastic support.

Locations:

- External ear (pinna).
- Epiglottis.
- Auditory (Eustachian) tube.
- Some small cartilages of larynx (e.g. cuneiform, corniculate).

D. Fibrocartilage



Structure:

- Dense bundles of type I collagen fibers.
- Fewer chondrocytes in lacunae (arranged in rows).
- Less ground substance than hyaline cartilage.

Function:

- Very strong; resists tensile forces and compression.
- Provides shock absorption.

Locations:

- Intervertebral discs.
- Pubic symphysis.
- Menisci of knee joint.
- Attachments of tendons and ligaments to bone.

E. Comparison of the three cartilage types

Feature	Hyaline cartilage	Elastic cartilage	Fibrocartilage
Main collagen type	Type II (fine fibers)	Type II (with elastic fibers)	Type I (thick bundles)
Elastic fibers	Absent	Abundant	Absent
Appearance (matrix)	Smooth, glassy	Many elastic fibers (wavy)	Dense, fibrous (bundles)
Chondrocytes	Many, often in groups	Many	Fewer, in rows
Ground substance	Abundant	Abundant	Less
Mechanical properties	Strong, resilient, low friction	Flexible, maintains shape	Very strong, resists tension and compression
Main locations	Articular cartilage, ribs, trachea, nose, fetal skeleton	External ear, epiglottis, Eustachian tube	Intervertebral discs, pubic symphysis, menisci, tendon/ligament attachments

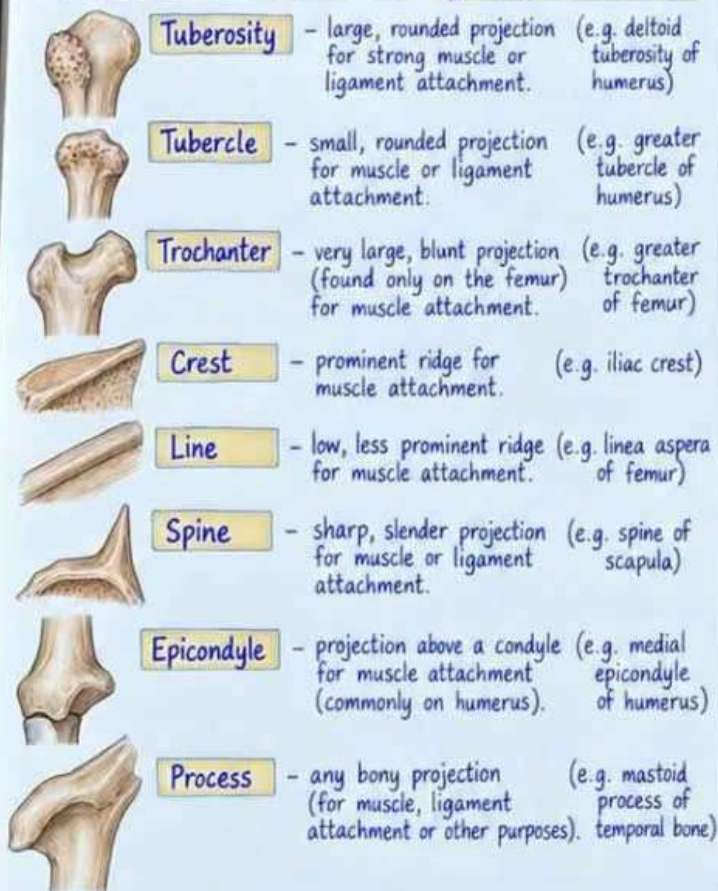
F. Functions and key points

- Cartilage is a type of connective tissue with chondrocytes in lacunae and a firm extracellular matrix.
- Provides support, flexibility and shock absorption.
- Avascular (no blood vessels), aneural (no nerves) and alymphatic (no lymphatics).
- Nutrients diffuse through the matrix from surrounding tissues (e.g. perichondrium or synovial fluid).
- Perichondrium (except at articular cartilage) supplies nutrients and contains chondroblasts for growth and repair.
- Three types: hyaline, elastic and fibrocartilage, each adapted to specific mechanical demands.
- Cartilage can grow by interstitial growth (division of chondrocytes) and appositional growth (from chondroblasts in perichondrium).
- Limited self-repair due to poor blood supply.

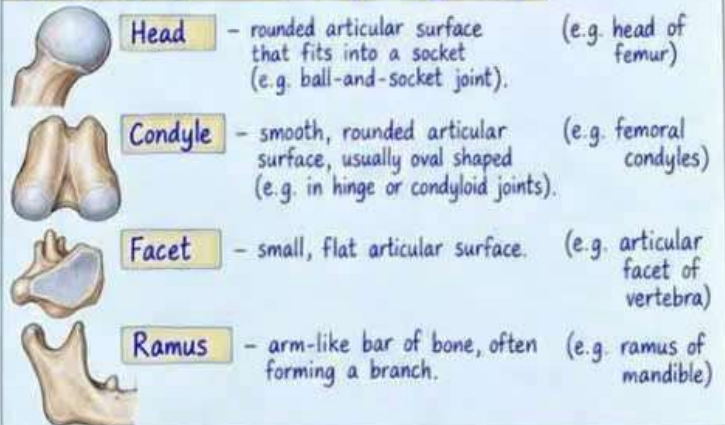
Flexible cartilage, strong support! ♥

BONE MARKINGS AND LANDMARKS

A. Projections for muscle or ligament attachment



B. Articular surfaces (form joints)



C. Depressions and openings



D. Example: Femur with labeled landmarks



E. Functions / why markings matter

- Provide attachment points for muscles and ligaments, enabling movement and joint stability.
- Form articular surfaces that allow bones to articulate and create joints.
- Allow passage of important structures such as nerves and blood vessels through foramina, canals and grooves.
- Help identify bones and their orientation in anatomy (important in study, imaging and surgery).
- Common clinical relevance:
 - Fractures often involve specific landmarks (e.g. femoral neck, tibial tuberosity).
 - Nerves or vessels can be compressed in grooves or canals (e.g. ulnar nerve in cubital tunnel).
 - Landmarks are used as reference points in procedures and injections (e.g. landmark-guided nerve blocks).

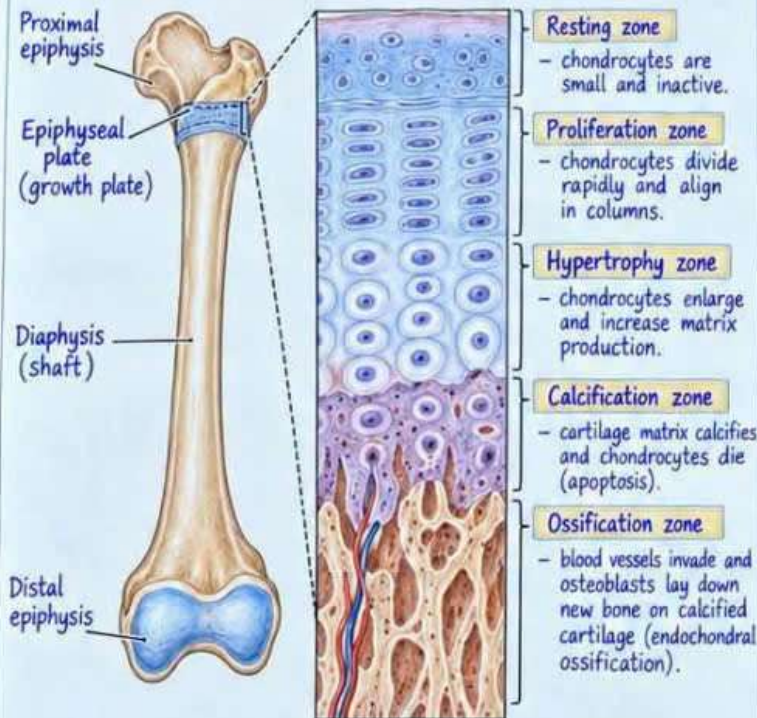
F. Summary table

Term	Meaning	Example
Tuberosity	large rounded projection	deltoid tuberosity
Tubercle	small rounded projection	greater tubercle (humerus)
Trochanter	very large projection (on femur)	greater trochanter
Crest	prominent ridge	iliac crest
Line	low ridge	linea aspera
Spine	sharp projection	spine of scapula
Epicondyle	projection above condyle	medial epicondyle
Process	any projection	mastoid process
Head	rounded articular surface	head of femur
Condyle	rounded articular surface	femoral condyle
Facet	small flat surface	articular facet
Ramus	arm-like bar	ramus of mandible
Fossa	shallow depression	olecranon fossa
Sulcus	narrow groove	bicipital sulcus
Foramen	round/oval opening	obturator foramen
Canal (meatus)	tube-like passage	vertebral canal
Fissure	narrow slit-like opening	superior orbital fissure
Sinus	air-filled cavity	maxillary sinus

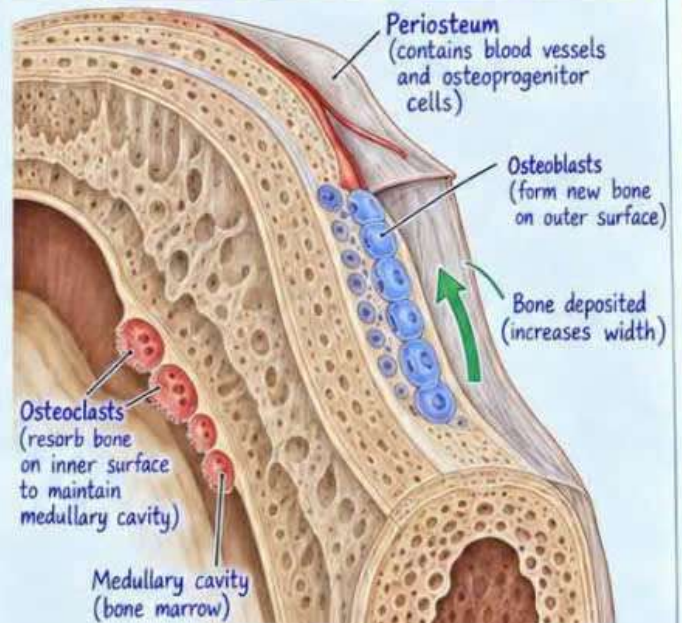
Landmarks give bones purpose! ❤️

BONE GROWTH AND REMODELING

A. Longitudinal growth at the epiphyseal plate

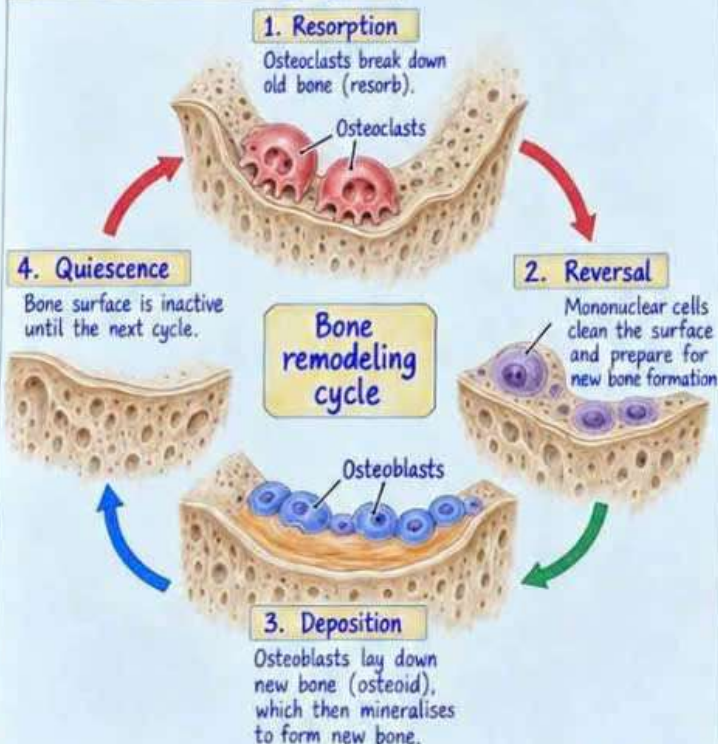


B. Appositional growth (increasing bone width)



- Appositional growth increases bone diameter (width).
- Occurs throughout life, helping to strengthen bones and adapt to mechanical load.

C. Bone remodeling cycle



- Remodeling removes old or damaged bone and replaces it with new bone.
- In adults, bone resorption and formation are balanced (coupled).
- Helps maintain bone strength, repair microdamage and regulate mineral levels (e.g. calcium and phosphate).

D. Factors affecting growth and remodeling

- **Growth hormone (GH)** – stimulates chondrocyte proliferation and increases bone growth (via IGF-1).
- **Thyroid hormone (T_3, T_4)** – essential for normal growth and bone maturation.
- **Sex hormones (oestrogen and testosterone)** – stimulate growth during puberty but eventually lead to closure of the epiphyseal plates.
- **Vitamin D** – increases calcium and phosphate absorption from the gut, essential for bone mineralisation.
- **Calcium** – provides the mineral for bone formation.
- **Parathyroid hormone (PTH)** – increases blood calcium (stimulates bone resorption, increases renal reabsorption of calcium, activates vitamin D).
- **Calcitonin** – inhibits osteoclast activity and reduces bone resorption (minor role).
- **Exercise / weight bearing** – mechanical load stimulates bone formation and increases bone density (Wolff's law).

E. Key points and summary

- Longitudinal growth occurs at the epiphyseal plates through endochondral ossification (resting → proliferation → hypertrophy → calcification → ossification).
- Appositional growth increases bone width by adding new bone to the outer surface (periosteal) and removing bone from the inner surface.
- Bone remodeling continuously replaces old bone with new bone (resorption followed by deposition).
- Growth and remodeling are regulated by hormones, nutrients and mechanical loading.
- The result is strong, healthy bones that can grow, adapt and repair throughout life.

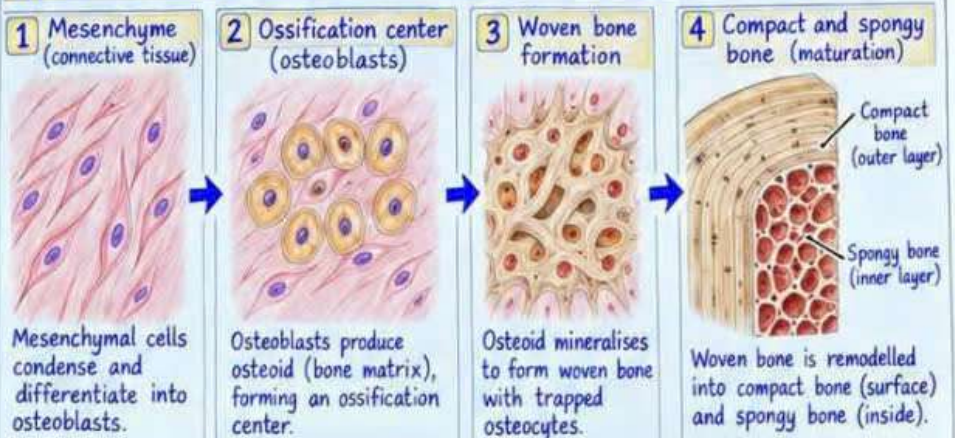
Healthy bones for a stronger future! ❤️

OSSIFICATION ANATOMY

A. Definition of ossification

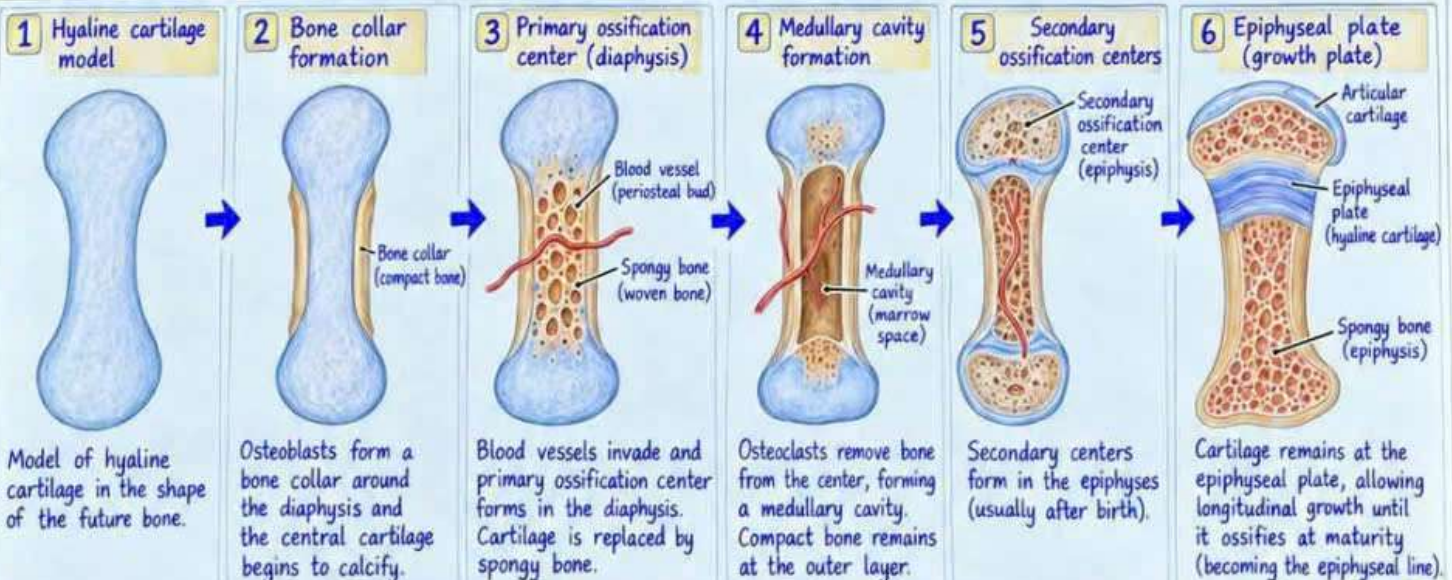
- Ossification (osteogenesis) is the process by which new bone tissue is formed.
- It occurs during fetal development, childhood, adolescence and also in fracture healing.
- Two main types:
 - Intramembranous ossification** - bone forms directly from mesenchyme.
 - Endochondral ossification** - bone replaces a cartilage model (hyaline cartilage).
- The result is formation of strong, living bone with osteocytes.

B. Intramembranous ossification (direct bone formation)



Examples Flat bones of the skull (e.g. frontal, parietal), mandible and clavicle.

C. Endochondral ossification (bone formation via cartilage model)



Examples Most long bones (e.g. femur, tibia, humerus), and also vertebrae, ribs and base of skull.

D. Comparison of the two processes

Feature	Intramembranous ossification	Endochondral ossification
Starting material	Mesenchyme	Hyaline cartilage model
Process	Direct bone formation	Cartilage replaced by bone
Main bones formed	Flat bones (skull, mandible, clavicle)	Long bones, vertebrae, ribs, base of skull
Growth pattern	Appositional growth	Length and width growth (via epiphyseal plate)
Speed	Relatively fast	Slower, more complex
Example	Frontal bone of skull	Femur

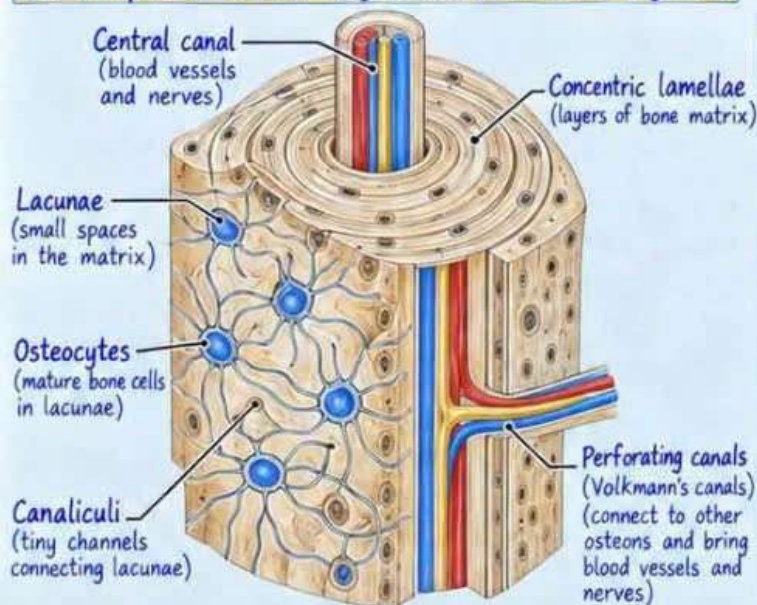
E. Clinical notes and key points

- Epiphyseal plate (growth plate) is hyaline cartilage where longitudinal growth occurs in long bones.
- When growth is complete (late adolescence), the epiphyseal plate ossifies and becomes the epiphyseal line.
- Disruption of the growth plate (e.g. injury) can lead to growth arrest and limb length discrepancies.
- During fetal development, most of the skeleton forms by endochondral ossification, while some bones (e.g. skull vault, clavicle) form by intramembranous ossification.
- Ossification continues after birth during growth and also plays a key role in fracture healing (both processes can occur).
- The final result is a strong, mineralised skeleton capable of supporting the body, protecting organs and enabling movement.

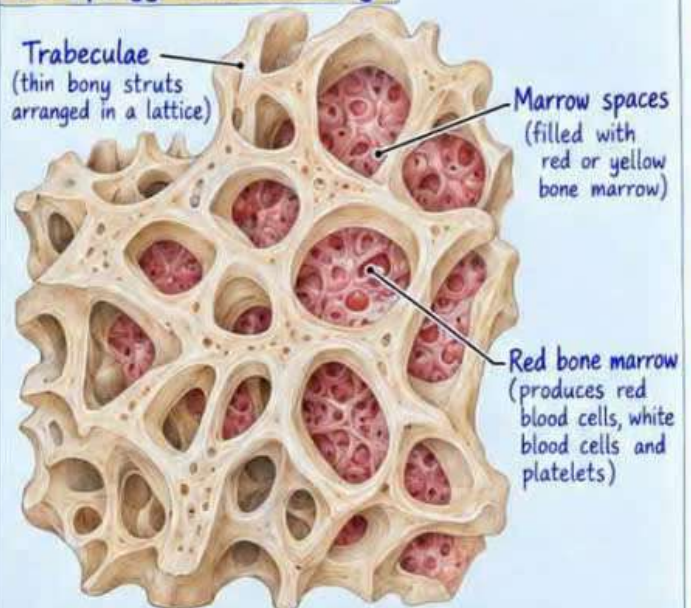
Bone today, stronger tomorrow! ❤️

COMPACT BONE VS SPONGY BONE

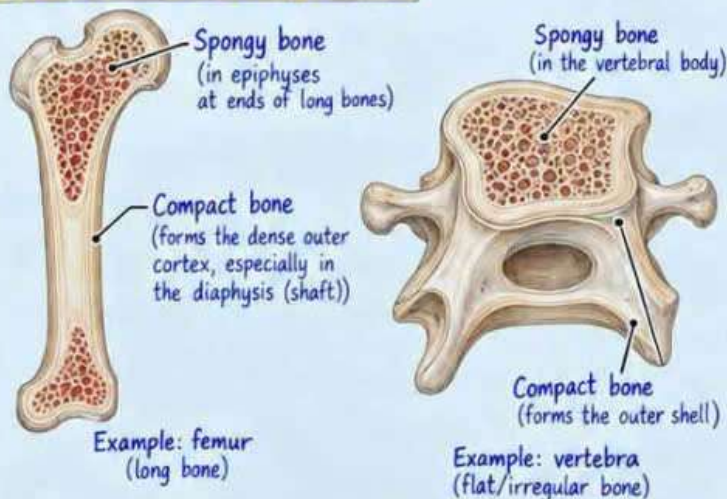
A. Compact bone anatomy (Osteon / Haversian system)



B. Spongy bone anatomy



C. Locations in the skeleton



D. Functions of compact and spongy bone

Compact bone

- Provides strength and rigidity.
- Resists bending, twisting and mechanical stress.
- Forms a hard outer shell that protects inner tissues.
- Provides a framework for muscle attachment.
- Contains osteons with blood vessels for nourishment.

Spongy bone

- Reduces the overall weight of the skeleton.
- Withstands forces from multiple directions (strong for its weight).
- Houses red bone marrow for blood cell production (hematopoiesis).
- Stores yellow bone marrow (fat) in some locations.
- Allows distribution of mechanical stress through trabeculae.

E. Side-by-side comparison

Feature	Compact bone	Spongy bone
Structure	Dense, solid and organized in concentric layers (lamellae)	Porous, lattice-like network of trabeculae
Density	High density	Lower density (more spaces)
Microscopic unit	Osteon (Haversian system)	Trabeculae
Main role	Strength and protection	Lightweight support and houses bone marrow
Common locations	Outer cortex of all bones, especially diaphysis of long bones	Epiphyses of long bones, inside flat bones (e.g. skull, ribs, sternum) and vertebrae

F. Key points / summary

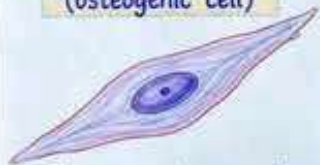
- Compact bone is dense, strong and forms the outer layer of bones.
- Spongy bone is porous, lighter and contains bone marrow.
- Compact bone is organized into osteons with a central canal, concentric lamellae, lacunae and canaliculi.
- Spongy bone consists of trabeculae with marrow spaces filled with red or yellow bone marrow.
- Both types of bone work together to make the skeleton strong, lightweight and functional.
- Compact bone resists bending and twisting, while spongy bone absorbs forces from multiple directions.
- Spongy bone is essential for blood cell production.
- Different bones contain different proportions of compact and spongy bone depending on their role.

Same bone, different structure, same amazing job! ♥

BONE CELLS AND BONE MATRIX

A. Bone cells overview

Osteoprogenitor cell (osteogenic cell)



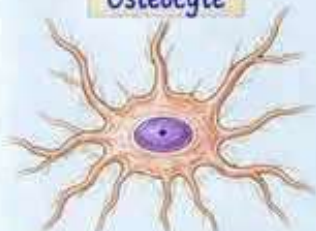
Undifferentiated stem cell that gives rise to osteoblasts (found in periosteum, endosteum and Haversian canals).

Osteoblast



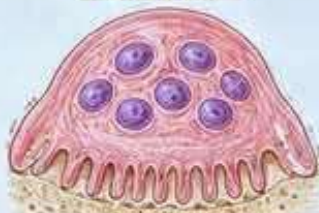
Form bone matrix (osteoid), secrete type I collagen and regulate mineralization.

Osteocyte



Mature bone cell trapped in a lacuna, with long processes that communicate through canaliculi.

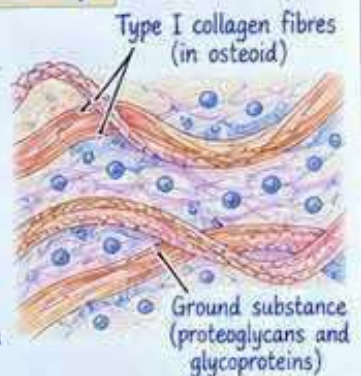
Osteoclast



Large multinucleated cell that resorbs bone by secreting acid and enzymes.

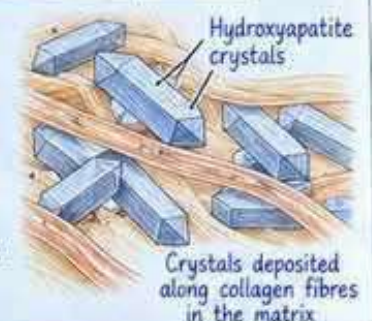
B. Organic matrix (osteoid)

- Mainly type I collagen fibres (about 90% of organic matrix).
- Ground substance containing proteoglycans (e.g. decorin, biglycan), glycoproteins (e.g. osteonectin, osteocalcin) and water.
- Provides a flexible scaffold for mineral deposition and gives bone its tensile strength and resistance to stretching.



C. Inorganic matrix (mineral component)

- Mainly hydroxyapatite crystals (calcium phosphate): $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$.
- Crystals are deposited within and around collagen fibres in the osteoid.
- Provides hardness, compressive strength and rigidity to bone.



D. Lacunae, canaliculi and matrix organization

Canaliculi (connects to other osteocytes and surfaces)

Osteocyte in lacuna

Concentric lamellae of bone matrix

- Osteocytes sit in small spaces called lacunae within the matrix.
- Long, thin processes extend through canaliculi to connect with other osteocytes, osteoblasts and bone surface.
- This network allows the exchange of nutrients, waste products and signalling molecules throughout bone.
- Organized into lamellae in compact bone (e.g. osteons) and a trabecular arrangement in spongy bone.

E. Functions of bone cells

Osteoblasts - bone formation

- Synthesize and secrete osteoid (type I collagen and ground substance).
- Initiate mineralization.
- Regulated by hormones and growth factors (e.g. PTH, vitamin D, IGF-1).

Osteocytes - maintenance

- Maintain bone matrix.
- Act as mechanosensors (detect mechanical load).
- Coordinate bone remodelling by signalling to osteoblasts and osteoclasts (e.g. via RANKL, sclerostin).

Osteoclasts - resorption

- Resorb bone by secreting H^+ (acid) and proteolytic enzymes (e.g. cathepsin K).
- Derived from monocyte/macrophage lineage.
- Activated by RANKL and inhibited by osteoprotegerin (OPG).

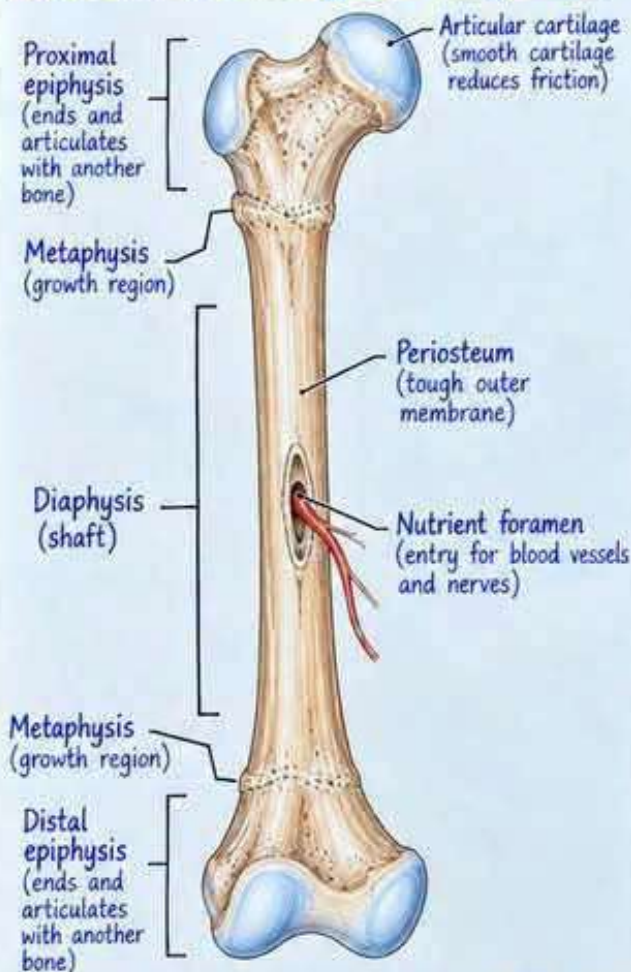
F. Summary of bone cells

Cell type	Appearance (key features)	Main location	Main function
Osteoprogenitor cell	Flattened, spindle-shaped	Periosteum, endosteum, Haversian canals	Stem cell; gives rise to osteoblasts (during growth and repair)
Osteoblast	Cuboidal, single nucleus, basophilic cytoplasm	Bone surface (periosteum, endosteum, trabeculae)	Synthesize osteoid and initiate mineralization
Osteocyte	Stellate with long processes	In lacunae within bone matrix	Maintain matrix; mechanosensor; coordinate remodelling
Osteoclast	Large, multinucleated, ruffled border	On bone surface in resorption bays (Howship's lacunae)	Resorb bone (secrete acid and enzymes)

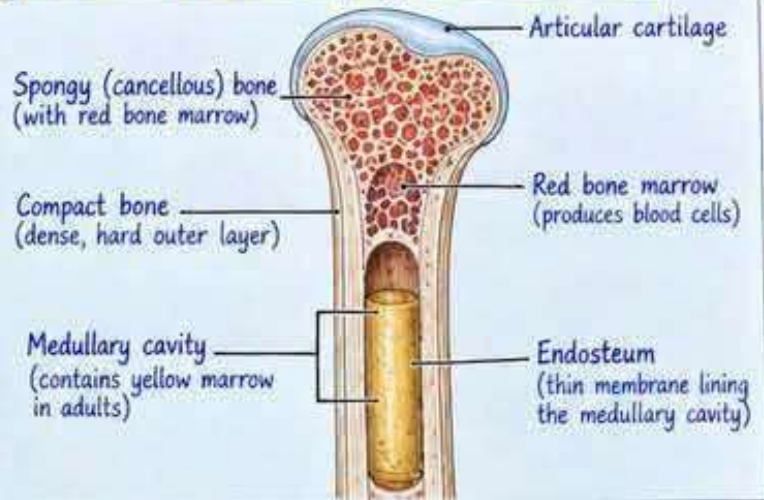
Different cells, one strong skeleton! ❤️

STRUCTURE OF A LONG BONE

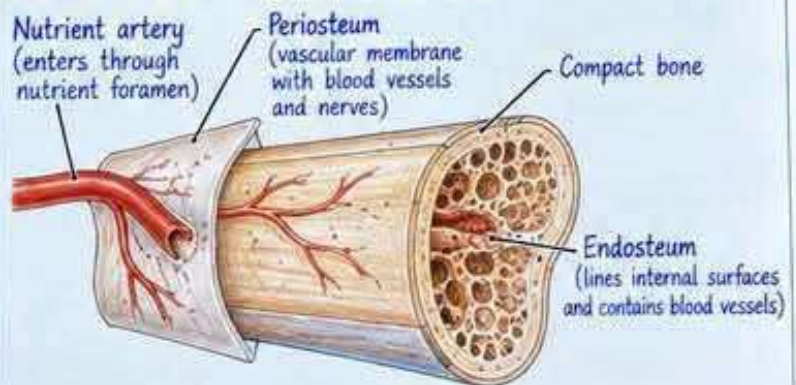
A. External anatomy of a long bone



B. Internal structure (longitudinal section)



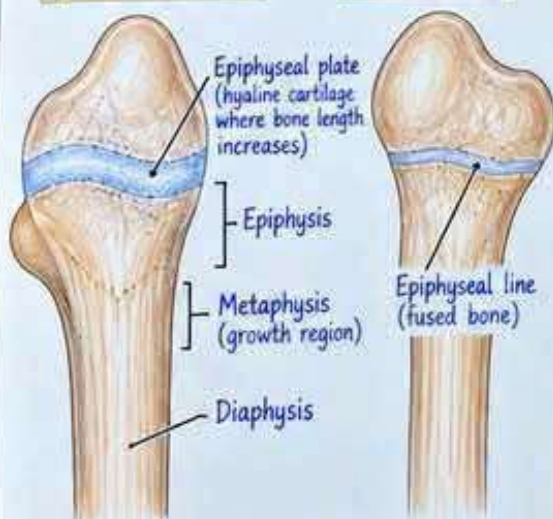
C. Blood supply and coverings



D. Epiphyseal plate / line (growth region)

During growth
(child / adolescent)

After growth
(adult)



E. Functions of the major parts

- **Articular cartilage** – reduces friction at joint surfaces.
- **Periosteum** – protects the bone, provides blood vessels and nerves, helps in growth and repair.
- **Compact bone** – strong outer layer for strength and protection.
- **Spongy bone** – lightweight structure with spaces, contains red marrow.
- **Medullary cavity** – houses yellow marrow (fat storage).
- **Endosteum** – lines internal surfaces, helps with bone remodelling.
- **Red bone marrow** – produces red blood cells, white blood cells and platelets.
- **Yellow bone marrow** – stores fat (energy reserve).

F. Key points / summary

- A long bone has a diaphysis (shaft), two epiphyses (ends) and metaphyses (growth regions).
- The outer surface is covered by a tough, vascular periosteum.
- Dense compact bone forms the outer layer, while spongy bone is found mainly in the epiphyses.
- The medullary cavity contains yellow bone marrow in adults.
- Red bone marrow is found in spongy bone and produces blood cells.
- Blood enters through nutrient foramina via nutrient arteries.
- The epiphyseal plate is a cartilaginous growth region which fuses to form an epiphyseal line in adults.
- Each part has a specific structure and function, working together for strength, movement, haematopoiesis and energy storage.

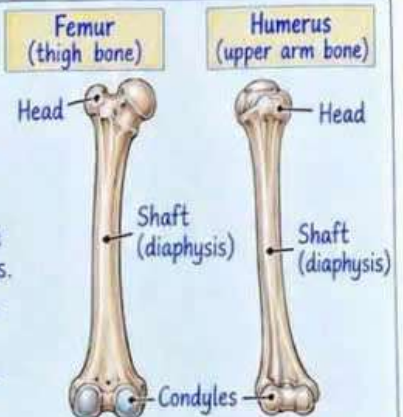
CLASSIFICATION OF BONES

A. Overview of bone classification by shape

- Bones can be classified into different types based on their shape.
- Each type has a structure adapted to its function in the body.
- The main types are long, short, flat, irregular, sesamoid and sutural bones.
- Although shapes differ, all bones are made of compact and spongy bone, and help to support, protect, enable movement, store minerals and produce blood cells.

B. Long bones

- Long bones are longer than they are wide.
- They have a shaft (diaphysis) with two ends (epiphyses).
- Mostly made of compact bone with spongy bone at the ends.
- Functions:** act as levers for movement, support body weight and contain red bone marrow.



C. Short bones

- Short bones are roughly cube-shaped (similar length, width and height).
- Made mostly of spongy bone with a thin outer layer of compact bone.
- Functions:** provide stability, allow small movements and withstand compression.



Example: carpal bones in the wrist



Example: tarsal bones in the ankle

D. Flat bones

- Flat bones are thin, flat and often curved.
- Made of two layers of compact bone with a layer of spongy bone (diploë) in between.
- Functions:** protect vital organs, provide broad surfaces for muscle attachment and offer large areas for blood cell production.



E. Irregular, sesamoid and sutural bones

Irregular bones



- Irregular bones have complex shapes that do not fit into other categories.
- Examples: vertebrae, hip bones, some bones of the face.

Sesamoid bones



- Sesamoid bones develop within tendons, usually near joints.
- They reduce friction, protect tendons and improve the efficiency of muscle contraction.
- Example: patella (kneecap).

Sutural bones



- Sutural bones are small, additional bones found within the sutures of the skull.
- They vary in number and are a normal anatomical variation.
- Example: wormian bones in the skull.

F. Functions and key points

- Bone shape is closely related to its function.
- All bone types contribute to support, protection, movement, mineral storage and blood cell formation.
- Long bones act as levers for movement.
- Short bones provide stability and resist compression.
- Flat bones protect vital organs and offer large areas for muscle attachment.
- Irregular bones have complex shapes for specialised roles.
- Sesamoid bones improve tendon function and reduce friction.
- Sutural bones are small extra bones in skull sutures.

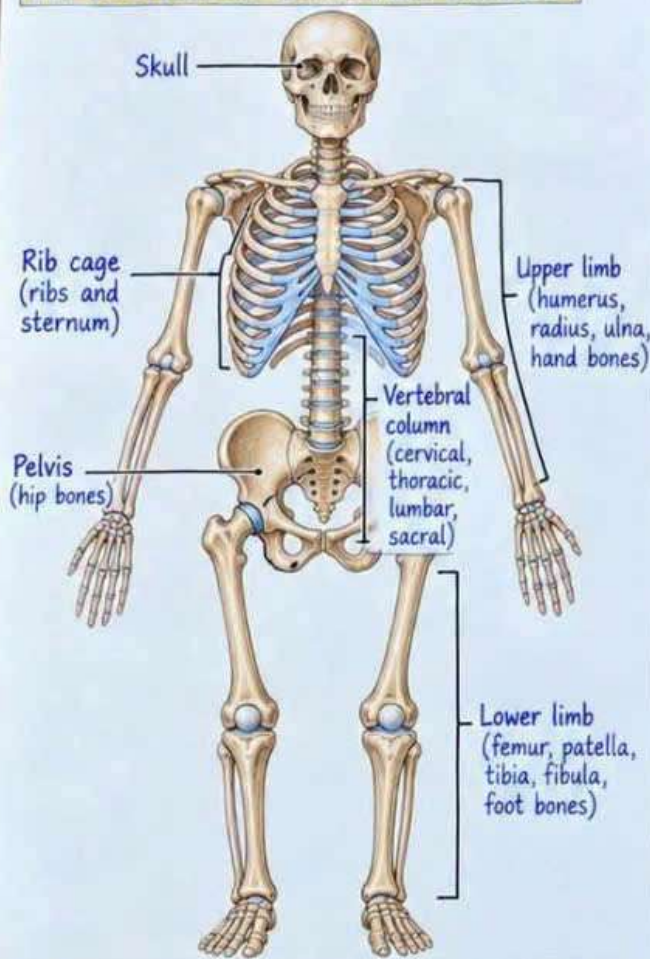
G. Summary table

Type of bone	Shape / characteristics	Main function(s)	Examples
Long bones	Longer than wide, with a shaft and two ends	Levers for movement, support, contain marrow	Femur, humerus
Short bones	Roughly cube-shaped	Stability, resist compression, allow small movements	Carpals, tarsals
Flat bones	Thin, flat, often curved	Protection, muscle attachment, blood cell production	Skull, sternum, ribs, scapula
Irregular bones	Complex shapes	Specialised functions (support, protection, etc.)	Vertebrae, hip bones, facial bones
Sesamoid bones	Small, rounded, within tendons	Reduce friction, protect tendons, improve muscle efficiency	Patella
Sutural bones	Small, extra bones in skull sutures	No major function, normal variation	Wormian bones

Different shapes, essential roles! ♥

FUNCTIONS OF THE SKELETAL SYSTEM

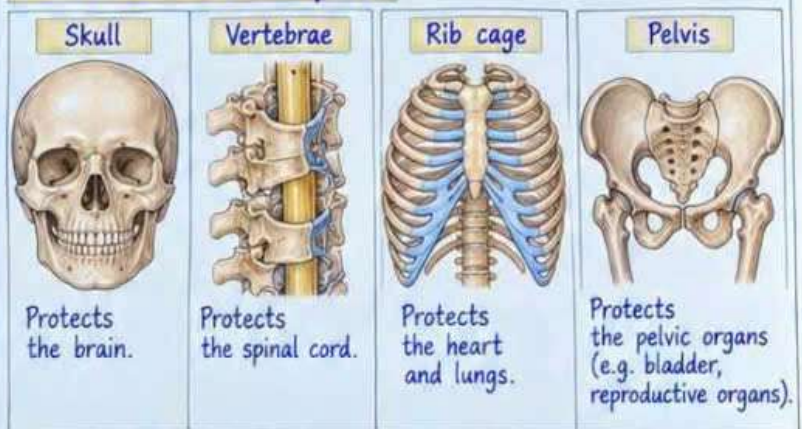
A. Overview of the human skeleton



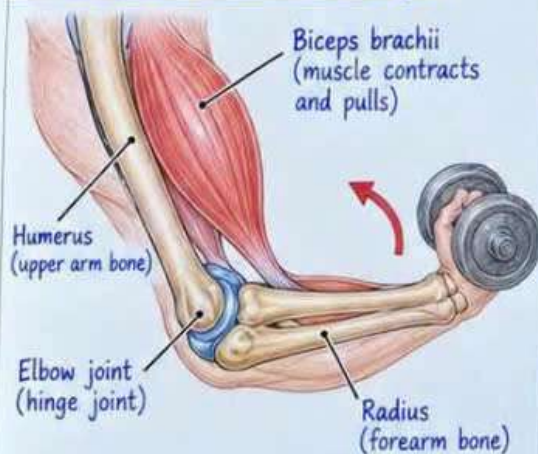
B. Main functions of the skeletal system

- **Support** - provides a rigid framework, gives the body shape and maintains posture.
- **Protection** - shields vital internal organs from mechanical injury.
- **Movement (leverage)** - bones act as levers and provide attachment points for muscles, enabling movement.
- **Mineral storage** - stores important minerals, especially calcium and phosphate, in the bone matrix.
- **Blood cell formation** - red bone marrow produces blood cells (hematopoiesis).
- **Fat storage** - yellow bone marrow stores fat (energy reserves).

C. Protection examples

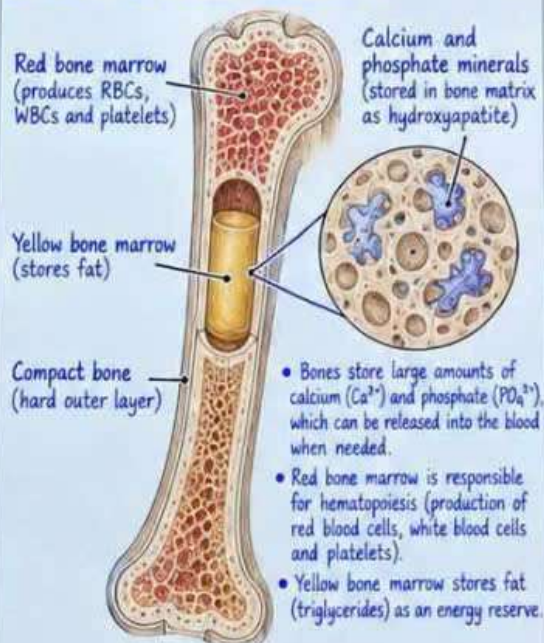


D. Movement and leverage



- Bones act as levers and muscles provide the force.
- When a muscle contracts, it pulls on a bone via a tendon, causing movement at a joint.
- The skeleton provides rigid attachment points for muscles and allows efficient movement.

E. Storage and formation



- Bones store large amounts of calcium (Ca^{2+}) and phosphate (PO_4^{3-}), which can be released into the blood when needed.
- Red bone marrow is responsible for hematopoiesis (production of red blood cells, white blood cells and platelets).
- Yellow bone marrow stores fat (triglycerides) as an energy reserve.

F. Key points / summary

- The skeletal system provides shape and a rigid framework for the body.
- Bones protect vital internal organs such as the brain, spinal cord, heart, lungs and pelvic organs.
- Bones act as levers and work with muscles to allow movement.
- Bone tissue stores important minerals (calcium and phosphate).
- Red bone marrow produces blood cells.
- Yellow bone marrow stores fat for energy.
- Overall, the skeletal system supports, protects, enables movement, stores minerals, produces blood cells and stores fat, all of which are essential for survival.

Strong bones, strong body! ♥